

Test Strategy Document

Project Name

Learning Management System (LMS)

Version

1.0

Prepared By

QA Engineer

Date

DD-MM-YYYY

1. Introduction

This Test Strategy document defines the testing approach, objectives, scope, testing types, test environment, risks, and deliverables for the Learning Management System (LMS).

The LMS application enables students to register, login, search courses, enroll in courses, track progress, download certificates, and allows instructors and administrators to manage learning content and users.

2. Testing Objectives

The primary objectives of testing are:

- Verify that all business requirements are implemented correctly.
 - Validate that the LMS functions as expected for students, instructors, and administrators.
 - Identify defects before deployment.
 - Ensure system stability and usability.
 - Verify data integrity and business rules.
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3. Scope of Testing

In Scope

Student Module

- Registration
- Login
- Forgot Password
- Course Search
- Course Enrollment
- Dashboard
- Progress Tracking
- Certificate Download

Instructor Module

- Course Creation
- Course Publishing

Admin Module

- User Management
 - User Activation/Deactivation
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Out of Scope

- Third-party payment gateway integration
 - Mobile application testing
 - Performance testing
 - Security penetration testing
 - Multi-language testing
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4. Requirements Covered

Requirement ID	Requirement Description
R1	Student Registration
R2	Student Login
R3	Forgot Password

R4	Course Search
R5	Course Enrollment
R6	Course Dashboard
R7	Course Progress Tracking
R8	Certificate Download
R9	Instructor Course Creation
R10	Admin User Management

5. Test Types

Functional Testing

Verify application functionality against business requirements.

Examples:

- Login validation
 - Course enrollment validation
 - Dashboard functionality
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Integration Testing

Verify interaction between modules.

Examples:

- Registration → Login
 - Enrollment → Dashboard
 - Progress Tracking → Certificate Generation
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Smoke Testing

Verify critical functionality after deployment.

Examples:

- Login
 - Enrollment
 - Dashboard Access
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Regression Testing

Verify that existing functionality works after bug fixes.

Examples:

- Login after password-related fixes
 - Enrollment after dashboard changes
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User Acceptance Support Testing

Support business users during UAT execution.

6. Test Design Techniques

The following test design techniques will be used:

- Equivalence Partitioning
 - Boundary Value Analysis
 - Decision Table Testing
 - Error Guessing
 - Positive Testing
 - Negative Testing
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7. Test Environment

Application Environment

- LMS Web Application
- Test Environment

Browser Coverage

- Google Chrome
- Microsoft Edge

Operating System

- Windows 10 / Windows 11
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8. Entry Criteria

Testing will begin when:

- Requirements are approved
 - Test environment is available
 - Build is deployed successfully
 - Test data is available
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9. Exit Criteria

Testing will be considered complete when:

- All planned test cases are executed
 - Critical defects are closed
 - High-priority defects are resolved or accepted
 - Test Summary Report is approved
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10. Defect Management Strategy

Defects will be logged and tracked using the defect management process.

Defect Lifecycle:

New → Assigned → Open → Fixed → Retest → Closed

Alternative statuses:

- Rejected
- Deferred
- Duplicate

Defects will be categorized based on:

- Severity
 - Priority
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11. Risks and Mitigation

Risk	Impact	Mitigation
Incomplete requirements	High	Requirement review meetings
Environment unavailability	High	Early environment validation
Delayed builds	Medium	Regular coordination with development team
Test data issues	Medium	Prepare test data in advance

12. Test Deliverables

The following deliverables will be produced:

- Requirement Clarification Log
 - Requirement Review Notes
 - Test Strategy
 - Test Plan
 - Test Scenarios
 - Test Cases
 - Test Data
 - RTM
 - Test Execution Report
 - Defect Log
 - Test Summary Report
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13. Approval

This Test Strategy document defines the overall testing approach for the LMS project and will be used as the baseline for all testing activities.